

Mahindra University Hyderabad
 École Centrale School of Engineering
 Minor-I

Program: B. Tech.

Branch: CSE/ECE/AI Year: III

Semester: I

Subject: Operating System (CS3102)

Date: 01/11/2025

Time Duration: 1.5 Hours

Start Time: 02:00 PM

Max. Marks: 30 Marks

Instructions:

1. Please write the answer in sequence as given in the question paper.
2. Attempt all questions.

Q. No.	Questions		
1	Explain how Test and Set is used for process synchronization. Explain with proper pseudocode and an example. (5 Marks)		
2	The reader-writer process synchronization pseudo-code is given below with some blanks. The process is synchronized using a mutex and a wrt semaphore. Write the proper word in place of the 10 blanks in the answer script. (10 Marks) <table border="1"> <tr> <td style="vertical-align: top;"> Reader : <pre> do { wait(1.-----); readcnt++; if (readcnt == 2.-----) wait(3.-----); signal(4.-----); Critical Section (Reading) wait(5.-----); readcnt--; if (readcnt == 6.-----) signal(7.-----); signal(8.-----); } while(true); </pre> </td><td style="vertical-align: top;"> Writer: <pre> do { wait(9.-----); Critical Section (Writing) signal(10.-----); } while(true); </pre> </td></tr> </table>	Reader : <pre> do { wait(1.-----); readcnt++; if (readcnt == 2.-----) wait(3.-----); signal(4.-----); Critical Section (Reading) wait(5.-----); readcnt--; if (readcnt == 6.-----) signal(7.-----); signal(8.-----); } while(true); </pre>	Writer: <pre> do { wait(9.-----); Critical Section (Writing) signal(10.-----); } while(true); </pre>
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3

A single-processor system has three resource types X, Y, and Z, which are shared by three processes. There are 5 units of each resource type. Consider the following scenario, where the column allocation denotes the number of units of each resource type allocated to each process, and the column request denotes the number of units of each resource type requested by a process in order to complete execution. Use the banker's algorithm to check if the system is in a safe state or not. If yes, write a safe sequence. **(5 Marks)**

	Allocation			Request		
	X	Y	Z	X	Y	Z
P0	1	2	1	1	0	3
P1	2	0	1	0	1	2
P2	2	2	1	1	2	0

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Suppose there are three identical printers shared among multiple processes. **(5 Marks)**

Design a semaphore-based solution to ensure that:

- No more than 3 processes can print simultaneously, and
 - Each process gets exclusive access to one printer at a time.
- Explain how your semaphores maintain correct resource usage.

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Two processes share two semaphores S1 and S2, both initialized to 1. **(5 Marks)**

Program-1

wait(S1);

wait(S2);

\\ Critical Section

Signal(S2)

Signal(S1)

Program-2

wait(S2);

wait(S1);

\\ Critical Section

Signal(S1)

Signal(S2)

- Illustrate how this system may enter a deadlock.
- Suggest one modification to prevent deadlock and explain why it works.